



Brief Paper

Multiple faults-tolerant control via fully actuated system approaches[☆]Miao Cai^{a,b,c}, Xiao He^c, Donghua Zhou^{a,b,c,*}^a School of Automation, Southeast University, Nanjing 210096, China^b Institute of Intelligent Unmanned Systems, Southeast University, Nanjing 210096, China^c Department of Automation, Tsinghua University, Beijing 100084, China

ARTICLE INFO

Article history:

Received 10 November 2024

Received in revised form 28 January 2026

Accepted 24 March 2026

Available online 2 May 2026

Keywords:

Fully actuated system approaches

Fault-tolerant control

Multiple faults

Observer saturation

ABSTRACT

The paper develops two multiple faults-tolerant control (MFTC) frameworks via fully actuated system (FAS) approaches. Starting from the observable space related to each measurement, the first MFTC structure provides a fusion estimation of high-order state and actuator faults. Under a fault detection logic based on the dynamical sensor index, the MFTC strategy can effectively accommodate unbounded sensor faults. In view of the noise effect on high-order systems, a saturated observer is redesigned in the second MFTC. One last thing, the proposed observer-saturated MFTC framework has better noise suppression performance. The system stability is theoretically proved and experimentally illustrated.

© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

Among the existing nonlinear control theories, the nonlinear first-order state-space theory has played an important role and spawned many well-known controller design methods. The state-space model has the advantage of a plain mathematical structure, which is beneficial for observer design and stability analysis. However, its complex control structure and difficult-to-tune parameters have limited the practical applicability of the state-space theory. With these limitations of state-space theory in mind, Duan creatively investigated another control theory – the high-order fully actuated system (HOFAS) theory (Duan, 2021).

Many common practical systems, e.g., drones, robotic arms, etc., are fully actuated systems (FASs) in the physical sense. Strict-feedback systems and feedback-linearized systems are all FASs in the mathematical sense (Duan, 2021). Obviously, most existing nonlinear systems have a certain degree of full-actuation characteristics in theoretical research and engineering experiments, but the research of these systems through state-space approaches is difficult to transfer each other (Duan, 2024).

To date, the HOFAS theory and application have been enriched to some extent. In Liu (2022), the HOFAS theory was successfully applied to the predictive control of networked multi-agent

systems. Furthermore, a formation control experiment of practical three air-bearing simulators illustrated the effectiveness of predictive control based on FAS approaches (Zhang et al., 2022). In Meng et al. (2022), the adaptive event-triggered control of FASs with unknown control coefficients was investigated. In Liu et al. (2023), an adaptive controller was designed for unknown mixed HOFASs without parameter overestimation. In Chen et al. (2023), a three-dimensional interception guiding law for large maneuvering objects was devised using FAS approaches, which can handle the terminal impact-angle constraint. In Cui et al. (2024), the trajectory tracking task of combined spacecraft simulators was completed through discrete-time FAS approaches. Besides, FAS approaches have also been successfully applied in many practical systems, e.g., quadrotors (Lu et al., 2023), multirotor aerial vehicles (Ricardo Jr & Santos, 2022), etc.

Not only the HOFAS theory under ideal conditions but also the HOFAS safety theory under nonideal conditions are flourishing. The fault-tolerant control (FTC) technology based on FAS approaches has received certain attention. In Cai et al. (2023a), an active fault tolerance strategy based on dynamical data models could effectively compensate catastrophic component faults. In Gong et al. (2024), a robust stabilization FAS controller was applied to the actuator fault tolerance. In Ma et al. (2023), a hierarchical prescribed-time controller was designed for multi-agent FASs against actuator faults. In Liu et al. (2024), the stabilization control of stochastic HOFASs with sensor faults was developed. In Dong et al. (2023), an adaptive FAS controller accommodated actuator faults under full-state constraints. In Cai et al. (2024), a redundant self-healing FTC technique tolerating unrestricted sensor faults was presented to reveal the essence of sensor fault tolerance. Actuator faults necessitate increased adaptive performance on controller design, and sensor faults can completely

[☆] This work was supported in part by the National Natural Science Foundation of China under Grant 62503111, Grant 62527805 and Grant U2541220, and in part by the Basic Research Program of Jiangsu under Grant BK20251256. The material in this paper was not presented at any conference. This paper was recommended for publication in revised form by Associate Editor Angelo Alessandri under the direction of Editor Thomas Parisini.

* Corresponding author at: School of Automation, Southeast University, Nanjing 210096, China.

E-mail addresses: caim25@seu.edu.cn (M. Cai), hexiao@tsinghua.edu.cn (X. He), zdth@tsinghua.edu.cn (D. Zhou).

ruin state measurements. In addition, measurement noise has a substantial impact on the observation and control of high-order systems. Although the previous deterministic FAS approaches cannot eliminate noise, the noise suppression performance of high-order systems needs some consideration.

Over the past years, there have been rapid development on FAS approaches, and a number of achievements have emerged in the FTC field based on this approaches. However, these previous works have the following limitations:

- (1) Most FTC technologies can only compensate a single type of actuator faults (Cai et al., 2023a; Dong et al., 2023; Gong et al., 2024; Liu et al., 2024; Ma et al., 2023), while few FTC technologies that can handle sensor faults cannot compensate actuator faults (Cai et al., 2024).
- (2) Most observer-based control instructions will oscillate due to noise. The higher the system order, the more obvious the oscillations will be.
- (3) Noise increases the difficulty of state observation and fault diagnosis for high-order systems.

Motivated by the aforementioned remarks, it is urgent to establish the multiple faults-tolerant control (MFTC) architecture via FAS approaches against two types of faults. The main contributions are as follows:

- (1) The fault-tolerant tracking control against both actuator and sensor faults is firstly completed via FAS approaches. Compared with the previous FTC techniques, the two MFTC frameworks based on observable decomposition have a greater tolerance range.
- (2) An observer-saturated MFTC strategy is redesigned to suppress measurement noise in high-order systems. Under this strategy, the fault diagnosis performance based on state observation will be further enhanced, and the accuracy and timeliness of control instructions will also be improved.

Notation: $\|\cdot\|$ denotes the vector norm or matrix norm. $\|\cdot\|_\infty$ means the infinite norm. $[m] \triangleq \{1, 2, \dots, m\}$ represents a sequence of natural numbers. For a set $\mathcal{I}$, $|\mathcal{I}|$ means the cardinality of it. $\text{Sat}_\pm(\cdot)$ means the saturation function. $\text{Lip}(\mathcal{Y})$, $\text{Lip}(\mathcal{Y})$ are the two bi-Lipschitz constants of a mapping $\mathcal{Y}$ satisfying $\text{Lip}(\mathcal{Y})\|z_1 - z_2\| \leq \|\mathcal{Y}(z_1) - \mathcal{Y}(z_2)\| \leq \overline{\text{Lip}}(\mathcal{Y})\|z_1 - z_2\|$. The other symbols are $x^{(0 \sim n-1)} = [x, \dot{x}, \dots, x^{(n-1)}]^\top$, $A_{0 \sim n-1} = [A_0, A_1, \dots, A_{n-1}]$, $B_n = [\mathbf{0}_{n-1}^\top, 1]^\top$, $C_n = [1, \mathbf{0}_{n-1}^\top]$,

$$\Phi_a = \begin{bmatrix} \mathbf{0}_{n-1} & I_{n-1} \\ -A_{0 \sim n-1} \end{bmatrix} \in \mathbb{R}^{n \times n}.$$

2. Problem formulation and preliminaries

Consider a class of nonlinear HOFASs with actuator faults and sensor faults:

$$\begin{aligned} \dot{x}^{(n)} &= F(x^{(0 \sim n-1)}) + G(x^{(0 \sim n-1)})u + \delta(t), \\ y_j &= H_j(x^{(0 \sim n-1)}) + s_j(t) + v_j, \quad j \in [m], \end{aligned} \quad (1)$$

where $x, u, y_j \in \mathbb{R}$ are the system state, the control input and the measurement output, respectively. The system state is in a compact set, i.e., $x^{(0 \sim n-1)} \in \Omega_x \triangleq \{\zeta \mid \zeta \in \mathbb{R}^n : \|\zeta\|_\infty \leq \mathcal{E}_x\}$ with a scalar $\mathcal{E}_x > 0$. The nonlinear functions F, G, H_j are smooth, and the controlled object (1) is fully actuated, i.e., $G(x^{(0 \sim n-1)}) \neq 0$, $\forall x^{(0 \sim n-1)} \in \Omega_x$ (Duan, 2021). There is measurement noise $v = [v_1, v_2, \dots, v_m]^\top$ with a tiny amplitude $\|v\|_\infty$. $\delta(t)$ is the possible actuator fault which can be modeled as (Cai et al., 2023b; Zhao et al., 2024)

$$\delta(t) = -\rho(t)G(x^{(0 \sim n-1)})u + f_a(t), \quad \rho(t) \neq 1, \quad (2)$$

where ρ, f_a are the time-varying drift fault and bias fault respectively. Sensor faults $s_j(t), j \in [m]$ occur randomly, and there are no limits on the boundedness and differentiability of $s_j(t)$. The ultimate research goal is the fault-tolerant tracking control design, i.e., to let x track a smooth reference signal x_r .

Assumption 1. The model dynamics and faults should meet the following conditions:

- (1) The known functions F, G, H_j are Lipschitz.
- (2) The reference signal satisfies $\sup_{t \geq 0} \|x_r^{(0 \sim n)}\| < \infty$.
- (3) The actuator fault satisfies $\sup_{t \geq 0} \|\delta, \dot{\delta}\| < \infty$.
- (4) There are at most d faulty measurements, and the amplitudes of sensor faults are possibly unbounded.

Property 1 (Duan, 2024). When the system (1) is assumed to be free of faults and noise, and full-order state $x^{(0 \sim n-1)}$ can be ideally measured, an FAS controller

$$u = -G^{-1}[A_{0 \sim n-1}e^{(0 \sim n-1)} + F + x_r^{(n)}] \quad (3)$$

can decouple a globally stable system $\dot{e}^{(0 \sim n-1)} = \Phi_a e^{(0 \sim n-1)}$ where $e^{(0 \sim n-1)} = x^{(0 \sim n-1)} - x_r^{(0 \sim n-1)}$, and Φ_a is a Hurwitz matrix.

However, the presence of faults and noise results in the inaccessibility of the controller (3). Traditional fault estimators via augmentation system approaches cannot compensate unconstrained sensor faults $s_j, j \in [m]$, because the traditional approaches require more stringent assumptions. Inspired by Cai et al. (2024), the redundant observability and m observable structures are analyzed as follows. Define $\mathcal{F} = [\dot{x}, \ddot{x}, \dots, x^{(n-1)}, F(x^{(0 \sim n-1)})]^\top$, $\mathcal{G} = [\mathbf{0}_{n-1}^\top, G(x^{(0 \sim n-1)})]^\top$, there exist m positive integers $r_j, j \in [m]$ such that $\forall t > 0, k \in [r_j - 2]$, $\mathcal{L}_G \mathcal{L}_{\mathcal{F}}^{r_j-1} H_j \neq 0$, $\mathcal{L}_G \mathcal{L}_{\mathcal{F}}^k H_j = 0$, where $\mathcal{L}$ denotes the Lie derivative. Define $\xi_j = [\xi_{j,1}, \xi_{j,2}, \dots, \xi_{j,r_j}]^\top = [\mathcal{L}_{\mathcal{F}} H_j, \mathcal{L}_{\mathcal{F}}^2 H_j, \dots, \mathcal{L}_{\mathcal{F}}^{r_j-1} H_j]^\top \triangleq \mathcal{Y}_j(x^{(0 \sim n-1)})$, there exists a diffeomorphism $[\xi_j^\top, \eta_j^\top]^\top = \mathfrak{h}_j(x^{(0 \sim n-1)})$ such that the j th subsystem is (Gauthier et al., 1992; Isidori, 1995)

$$\dot{\xi}_j = \begin{bmatrix} \xi_{j,1} \\ \xi_{j,2} \\ \vdots \\ \xi_{j,r_j} \end{bmatrix} = \begin{bmatrix} \xi_{j,2} \\ \vdots \\ \xi_{j,r_j} \\ F_o^j(\xi_j) + \delta \end{bmatrix} + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ G_o^j(\xi_j) \end{bmatrix} u, \quad (4a)$$

$$\dot{\eta}_j = F_{uo}^j(\xi_j, \eta_j) + G_{uo}^j(\xi_j, \eta_j)u + K_{uo}^j(\xi_j, \eta_j)\delta, \quad (4b)$$

$$y_j = \xi_{j,1} + s_j + v_j, \quad (4c)$$

where $F_o^j, G_o^j, F_{uo}^j, G_{uo}^j, K_{uo}^j$ are the Lipschitz nonlinear functions and $\xi_j, j \in [m]$ are in compact sets, i.e., $\xi_j \in \Omega_j \triangleq \{\zeta \mid \zeta \in \mathbb{R}^{r_j} : \|\zeta\|_\infty \leq \mathcal{E}_j\}$ with $\mathcal{E}_j > 0$. ξ is the overall observable structure of (1) where $\xi = [\xi_1^\top, \xi_2^\top, \dots, \xi_m^\top]^\top \triangleq \mathcal{Y}_{[m]}(x^{(0 \sim n-1)})$.

Property 2 (Cai et al., 2024). The HOFAS model (1) is uniformly observable if the mapping $\mathcal{Y}_{[m]}$ is an injective immersion. Consider a mapping $\mathcal{Y}_I, \forall I \subset [m], |I| = m - d$,

$$\xi_I = [\xi_{I,1}^\top, \xi_{I,2}^\top, \dots, \xi_{I,m-d}^\top]^\top \triangleq \mathcal{Y}_I(x^{(0 \sim n-1)}), \quad (5)$$

the HOFAS model (1) is d -redundant observable if each mapping $\mathcal{Y}_I, I \subset [m]$ is an injective immersion.

3. MFTC design via observer-extended FAS approaches

In the HOFAS model (1) with both actuator and sensor faults, the capability of passive FTC (PFTC) without fault diagnosis units is limited. As a result, the active FTC (AFTC) strategy is applied into the MFTC design. The residual signal, as an indicator based on state observation for fault diagnosis, is a crucial step in AFTC. The following research is divided into two parts.

3.1. FTC design without sensor faults

For measurement y_j , a single observer is designed as

$$\begin{aligned}\dot{\hat{\xi}}_{j,k} &= \hat{\xi}_{j,k+1} + \ell^k L_{j,k}(y_j - \hat{\xi}_{j,1}), k \in [r_j - 1], \\ \dot{\hat{\xi}}_{j,r_j} &= F_o^j(\hat{\xi}_j) + \hat{\delta}_j + \ell^{r_j} L_{j,r_j}(y_j - \hat{\xi}_{j,1}) + G_o^j(\hat{\xi}_j)u, \\ \dot{\hat{\delta}}_j &= \ell^{r_j+1} L_{j,r_j+1}(y_j - \hat{\xi}_{j,1}),\end{aligned}\quad (6)$$

where $\ell > 1$, $\hat{\xi}_j = [\hat{\xi}_{j,1}, \hat{\xi}_{j,2}, \dots, \hat{\xi}_{j,r_j}]^T$ is the observation of $\xi_j = [\xi_{j,1}, \xi_{j,2}, \dots, \xi_{j,r_j}]^T$, and $\hat{\delta}_j$ is the actuator fault estimation under the measurement y_j . $L_j = [L_{j,1}, L_{j,2}, \dots, L_{j,r_j+1}]$ should shape a Hurwitz matrix

$$\Psi_j = \begin{bmatrix} -L_j & \\ -I_{r_j} & \mathbf{0}_{r_j-1} \end{bmatrix}^T. \quad (7)$$

The unified observation is $\hat{\xi} = [\hat{\xi}_1^T, \hat{\xi}_2^T, \dots, \hat{\xi}_m^T]^T$. When (1) is observable, i.e., the mapping $\Upsilon_{[m]}$ is invertible, the observation $\hat{x}^{(0 \sim n-1)}$ can be calculated as

$$\begin{aligned}\hat{x}^{(0 \sim n-1)} &= \text{INV}(\Upsilon_{[m]})(\hat{\xi}) \\ &\triangleq \text{Sat}_{\mathcal{E}_x} \left(\Upsilon_{[m]}^{-1} \left([\text{Sat}_{\mathcal{E}_1} \hat{\xi}_1, \dots, \text{Sat}_{\mathcal{E}_m} \hat{\xi}_m]^T \right) \right),\end{aligned}\quad (8)$$

where $\Upsilon_{[m]}^{-1}$ is the inverse mapping of $\Upsilon_{[m]}$ in compatible compact sets (Shim, 2000). The unified fault estimation of δ is $\hat{\delta} = \sum_{j \in [m]} \hat{\delta}_j/m$. Then, the observer-extended controller tolerating actuator faults is

$$\begin{aligned}u_1 &= -G^{-1}(\hat{x}^{(0 \sim n-1)})[A_{0 \sim n-1}(\hat{x}^{(0 \sim n-1)} - x_r^{(0 \sim n-1)}) \\ &\quad + F(\hat{x}^{(0 \sim n-1)}) + \hat{\delta} - x_r^{(n)}].\end{aligned}\quad (9)$$

Lemma 1 (Khalil, 2017). For the observable system (1) with only actuator faults under the FTC (6)–(9), the overall error is ultimately uniformly bounded (UUB). Thus, there exist positive scalars $t_1, t_2, M_k (k \in [6])$ such that

$$\begin{aligned}\|x^{(0 \sim n-1)} - \hat{x}^{(0 \sim n-1)}\| &\leq M_1/\ell^2 + \ell^{n-1} M_2 \|v\|_\infty, t > t_1, \\ \|\delta - \hat{\delta}\| &\leq M_3/\ell + \ell^n M_4 \|v\|_\infty, t > t_1, \\ \|e^{(0 \sim n-1)}\| &\leq M_5/\ell + \ell^n M_6 \|v\|_\infty, t > t_2.\end{aligned}\quad (10)$$

Remark 1. The high-order states are obtained through observers, and the observers (6) have the stepwise error transmission structures. The ultimate error is principally influenced by the parameter ℓ , the system order n and the noise amplitude $\|v\|_\infty$.

3.2. MFTC design with multiple faults

The AFTC, including the fault detection logic, is the focus of this subsection. In order to deduce the residual signal, the state observation error of (4) with actuator faults and sensor faults is $\xi - \hat{\xi} = \epsilon + \tau_{[m]}$. ϵ is the error uncorrelated with sensor faults, and $\tau_{[m]}$ is the error stemming from only sensor faults. Based on Lemma 1, the error ϵ satisfies $\|\epsilon\| \leq \gamma \triangleq \sum_{j \in [m]} \max\{\lambda_j e^{-\alpha_j t}, M_1/\ell^2 + \ell^{n-1} M_2 \|v\|_\infty\}$. In the ideal sensor set U , τ_U equals zero.

Lemma 2 (Cai et al., 2024; Kim et al., 2019; Lee et al., 2019). For the system (1) with at most d sensor faults, the m measurements can determine $x^{(0 \sim n-1)}$ if and only if (1) is 2d-redundant observable.

When the 2d-redundant observable system (1) encounters at most d sensor faults, a fault residual signal based on the sensor set I is $\mathfrak{R}_I = \Upsilon_I(\text{INV}(\Upsilon_I)(\hat{\xi}_I)) - \hat{\xi}_I$, $|I| = m - d$. The definition of $\text{INV}(\Upsilon_I)$ is similar to (8).

Lemma 3 (Cai et al., 2024). There exist sensor faults in the set I if

$$\|\mathfrak{R}_I\| > \overline{\text{Lip}}(\Upsilon_I \circ \text{INV}(\Upsilon_I) - \mathcal{I})\gamma, \quad (11)$$

where $\circ$ means the composition of mappings, and $\mathcal{I}$ is the identity mapping. Conversely, if (11) fails in the 2d-redundant observable system (1) with no more than d sensor faults, the state observation, actuator fault estimation and sensor fault estimations are designed as

$$\begin{aligned}\hat{X}_I &\triangleq \hat{x}_I^{(0 \sim n-1)} = \text{INV}(\Upsilon_I)(\hat{\xi}_I), \hat{\delta}_I = \sum_{j \in I} \hat{\delta}_j/(m - d), \\ \hat{s}_{j,I} &= y_j - H_j(\hat{x}_I^{(0 \sim n-1)}), j \in [m].\end{aligned}\quad (12)$$

Thus, there exist positive scalars $t_3, \mathcal{M}_1, \mathcal{M}_2$ such that

$$\|x^{(0 \sim n-1)} - \hat{X}_I\| \leq \frac{\mathcal{M}_1}{\ell^2} + \ell^{n-1} \mathcal{M}_2 \|v\|_\infty, t > t_3. \quad (13)$$

Based on Lemma 3, the dynamical sensor set $\sigma(t)$, satisfying $|\sigma(t)| = m - d$, is updated as the next set $\sigma(t^+)$ if

$$\|\mathfrak{R}_{\sigma(t)}\| > \overline{\text{Lip}}(\Upsilon_{\sigma(t)} \circ \text{INV}(\Upsilon_{\sigma(t)}) - \mathcal{I})\gamma, \quad (14)$$

where t^+ represents the next moment in computer calculation, and the update of $\sigma(t)$ is ergodic. Under m observers (6), the traditional MFTC is designed as

$$\begin{aligned}\hat{x}_{\sigma(t)}^{(0 \sim n-1)} &= \text{INV}(\Upsilon_{\sigma(t)})(\hat{\xi}_{\sigma(t)}), \hat{\delta}_{\sigma(t)} = \sum_{j \in \sigma(t)} \hat{\delta}_j/(m - d), \\ \hat{s}_{j,\sigma(t)} &= y_j - H_j(\hat{x}_{\sigma(t)}^{(0 \sim n-1)}), j \in [m], \\ u_2 &= -G^{-1}(\hat{x}_{\sigma(t)}^{(0 \sim n-1)})[A_{0 \sim n-1}(\hat{x}_{\sigma(t)}^{(0 \sim n-1)} - x_r^{(0 \sim n-1)}) \\ &\quad + F(\hat{x}_{\sigma(t)}^{(0 \sim n-1)}) + \hat{\delta}_{\sigma(t)} - x_r^{(n)}],\end{aligned}\quad (15)$$

where $\hat{x}_{\sigma(t)}^{(0 \sim n-1)}, \hat{\delta}_{\sigma(t)}, \hat{s}_{j,\sigma(t)}$ are signal estimations.

Lemma 4. For the 2d-redundant observable system (1) with multiple faults under the MFTC (15), the overall error is UUB. Thus, there exist positive scalars $t_4, t_5, M_k (k = 7, 8, \dots, 14)$ such that

$$\begin{aligned}\|x^{(0 \sim n-1)} - \hat{x}_{\sigma(t)}^{(0 \sim n-1)}\| &\leq M_7/\ell^2 + \ell^{n-1} M_8 \|v\|_\infty, t > t_4, \\ \|\delta - \hat{\delta}_{\sigma(t)}\| &\leq M_9/\ell + \ell^n M_{10} \|v\|_\infty, t > t_4, \\ \|s_j - \hat{s}_{j,\sigma(t)}\| &\leq M_{11}/\ell^2 + \ell^{n-1} M_{12} \|v\|_\infty, j \in [m], t > t_4, \\ \|e^{(0 \sim n-1)}\| &\leq M_{13}/\ell + \ell^n M_{14} \|v\|_\infty, t > t_5.\end{aligned}\quad (16)$$

Proof. The proof is omitted. ■

4. MFTC redesign via observer-saturated FAS approaches

4.1. FTC redesign without sensor faults

To reduce the effect of $\ell^n \|v\|_\infty$, m observers with dynamical saturation levels $\tau_j, j \in [m]$ are redesigned as

$$\begin{aligned}\dot{\xi}'_{j,k} &= \xi'_{j,k+1} + L_{j,k} \text{Sat}_{\sqrt{\tau_j}}(\vartheta_j)/\ell^{r_j-k}, k \in [r_j - 1], \\ \dot{\xi}'_{j,r_j} &= F_o^j(\xi'_j) + \hat{\delta}'_j + G_o^j(\xi'_j)u + L_{j,r_j} \text{Sat}_{\sqrt{\tau_j}}(\vartheta_j), \\ \dot{\delta}'_j &= \ell L_{j,r_j+1} \text{Sat}_{\sqrt{\tau_j}}(\vartheta_j), \\ \dot{\tau}_j &= -\ell \theta \tau_j + \ell(\theta + \kappa) \vartheta_j^2, \tau_j \geq 0,\end{aligned}\quad (17)$$

where $\theta, \kappa > 0$, $\xi'_j = [\xi'_{j,1}, \xi'_{j,2}, \dots, \xi'_{j,r_j}]^T$, $\vartheta_j = \ell^{r_j}(y_j - \xi'_{j,1})$. The fused observation $\hat{x}^{(0 \sim n-1)'} is$

$$\begin{aligned}\hat{x}^{(0 \sim n-1)'} &= \text{INV}(\Upsilon_{[m]})(\hat{\xi}') \\ &\triangleq \text{Sat}_{\mathcal{E}_x} \left(\Upsilon_{[m]}^{-1} \left([\text{Sat}_{\mathcal{E}_1} \hat{\xi}'_1, \dots, \text{Sat}_{\mathcal{E}_m} \hat{\xi}'_m]^T \right) \right).\end{aligned}\quad (18)$$

The unified fault estimation is $\hat{\delta}' = \sum_{j \in [m]} \hat{\delta}'_j / m$. Hence, the observer-saturated controller is redesigned as

$$u'_1 = -G^{-1}(\hat{x}^{(0 \sim n-1)'})[A_{0 \sim n-1}(\hat{x}^{(0 \sim n-1)'}) - x_r^{(0 \sim n-1)}] + F(\hat{x}^{(0 \sim n-1)'}) + \hat{\delta}' - x_r^{(n)}. \quad (19)$$

Lemma 5 (Astolfi et al., 2021). For the observable system (1) with only actuator faults under the FTC (17)–(19), the overall error is UUB. Thus, there exist positive scalars $t'_1, t'_2, M'_k (k \in [6])$ such that

$$\begin{aligned} \|\hat{x}^{(0 \sim n-1)} - \hat{x}^{(0 \sim n-1)'}\| &\leq M'_1/\ell^2 + \ell^{n-1}M'_2\|v\|_\infty, t > t'_1, \\ \|\delta - \hat{\delta}'\| &\leq M'_3/\ell + \ell^n M'_4\|v\|_\infty, t > t'_1, \\ \|\hat{e}^{(0 \sim n-1)}\| &\leq M'_5/\ell + \ell^n M'_6\|v\|_\infty, t > t'_2. \end{aligned} \quad (20)$$

Proof. The estimation error for y_j is $\zeta'_j = [\zeta'_{j,1}, \dots, \zeta'_{j,r_j}, \zeta'_{j,r_j+1}]^T = [\ell^{\tau_j}(\xi_{j,1} - \hat{\xi}'_{j,1}), \dots, \delta - \hat{\delta}']^T$, and the derivatives of ζ'_j, τ_j are

$$\begin{aligned} \dot{\zeta}'_{j,k} &= \ell \zeta'_{j,k+1} - \ell L_{j,k} \zeta'_{j,1} + \ell L_{j,k} q_j + \ell L_{j,k} Q_j, k \in [r_j - 1], \\ \dot{\zeta}'_{j,r_j} &= \ell \zeta'_{j,r_j+1} - \ell L_{j,r_j} \zeta'_{j,1} + \ell L_{j,r_j} q_j + \ell L_{j,r_j} Q_j + \varpi'_j, \\ \dot{\zeta}'_{j,r_j+1} &= \delta - \ell L_{j,r_j+1} \zeta'_{j,1} + \ell L_{j,r_j+1} q_j + \ell L_{j,r_j+1} Q_j, \\ \dot{\tau}_j &= -\ell \theta \tau_j + \ell(\theta + \kappa) \zeta'^2_{j,2} \\ &\quad + \ell(\theta + \kappa)(\ell^{2\tau_j} v_j^2 + 2\ell^{\tau_j} v_j \zeta'_{j,1}), \end{aligned} \quad (21)$$

where $q_j = \zeta'_{j,1} - \text{Sat}_{\tau_j}(\zeta'_{j,1})$, $Q_j = \text{Sat}_{\tau_j}(\zeta'_{j,1}) - \text{Sat}_{\tau_j}(\zeta'_{j,1} + \ell^{\tau_j} v_j)$, $\varpi'_j = \ell[F^j_0(\xi_j) - F^j_0(\hat{\xi}'_j) + (G^j_0(\xi_j) - G^j_0(\hat{\xi}'_j))u'_1]$. It turns out that $\|q_j\| \leq \|\zeta'_{j,1}\|$, $\|Q_j\| \leq \|\ell^{\tau_j} v_j\| \leq \ell^{\tau_j} \|v\|_\infty$, $\|\varpi'_j\|, \|\varrho'_1\| \leq a'_1 + a'_2 \|\zeta'\| + a'_3 \|e^{(0 \sim n-1)}\|$ where a'_1, a'_2, a'_3 are positive scalars. The tracking error $e^{(0 \sim n-1)} = [e_1, e_2, \dots, e_n]^T$ satisfies $\dot{e}_i = e_{i+1}, i \in [n-1], \dot{e}_n = \varrho'_1 + A_{0 \sim n-1} \varepsilon' - A_{0 \sim n-1} e^{(0 \sim n-1)}$ where $\varepsilon' = x^{(0 \sim n-1)} - \hat{x}^{(0 \sim n-1)'}$, $\varrho'_1 = F(x^{(0 \sim n-1)}) - F(\hat{x}^{(0 \sim n-1)'}) + \delta - \hat{\delta} + [G(x^{(0 \sim n-1)}) - G(\hat{x}^{(0 \sim n-1)'})]u'_1$. A Lyapunov function is chosen as

$$V'_1 = \sum_{j=1}^m \zeta'^{\tau_j} P_j \zeta'_j + v_j \tau_j + \mu_j \max\{\zeta'^2_{j,1} - \tau_j, 0\} + e^{(0 \sim n-1)\tau} \mathcal{P} e^{(0 \sim n-1)}, \quad (22)$$

where μ_j, v_j are positive scalars satisfying $\mu_j > v_j$. P_j derives from $P_j \Psi_j + \Psi_j^T P_j = -I_{r_j+1}$ and $\mathcal{P}$ is the positive definite solution to $\mathcal{P} \Phi_a + \Phi_a^T \mathcal{P} = -\kappa I_n, \kappa > 0$. The upper Dini derivative is defined as $D^+ V'_1 \triangleq \lim_{h \rightarrow 0^+} (V'_1(t+h) - V'_1(t))/h$. Let $\mathfrak{V}_j = \zeta'^{\tau_j} P_j \zeta'_j + v_j \tau_j + \mu_j \max\{\zeta'^2_{j,1} - \tau_j, 0\}$ and $V'_1 = \sum_{j=1}^m \mathfrak{V}_j + e^{(0 \sim n-1)\tau} \mathcal{P} e^{(0 \sim n-1)}$, and the Dini derivative of $\mathfrak{V}_j$ can be divided into two cases.

Case 1: $\zeta'^2_{j,1} < \tau_j$. The Dini derivative of $\mathfrak{V}_j$ is

$$\begin{aligned} D^+ \mathfrak{V}_j &= \ell \zeta'^{\tau_j} (P_j \Psi_j + \Psi_j^T P_j) \zeta'_j + 2\ell Q_j L_j P_j \zeta'_j \\ &\quad + 2\delta B_{r_j+1}^T P_j \zeta'_j + 2\varpi'_j \mathfrak{B}_j^T P_j \zeta'_j + v_j [-\ell \theta \tau_j \\ &\quad + \ell(\theta + \kappa) \zeta'^2_{j,2} + \ell(\theta + \kappa)(\ell^{2\tau_j} v_j^2 + 2\ell^{\tau_j} v_j \zeta'_{j,1})], \end{aligned} \quad (23)$$

where $\mathfrak{B}_j = [B_{r_j}^T, 0]^T$. Then, there exist positive scalars a'_4, a'_5 such that

$$\begin{aligned} D^+ \mathfrak{V}_j &< -\ell[1 - v_j(\theta + \kappa)] \|\zeta'_j\|^2 + \ell^{\tau_j+1} a'_4 \|v\|_\infty \|\zeta'_j\| \\ &\quad + a'_5 \|\zeta'_j\| (a'_1 + a'_2 \|\zeta'\| + a'_3 \|e^{(0 \sim n-1)}\|) \\ &\quad - \ell v_j \theta \tau_j + \ell^{2\tau_j+1} v_j (\theta + \kappa) \|v\|_\infty^2 \\ &< -\ell[0.5 - v_j(\theta + \kappa) - a'_2 a'_5 / \ell] \|\zeta'_j\|^2 + (\ell^{\tau_j+1} a'_4 \|v\|_\infty \\ &\quad + a'_1 a'_5) \|\zeta'_j\| + a'^2_2 a'^2_5 \|e^{(0 \sim n-1)}\|^2 / (2\ell) \\ &\quad + \ell^{2\tau_j+1} v_j (\theta + \kappa) \|v\|_\infty^2. \end{aligned} \quad (24)$$

Case 2: $\zeta'^2_{j,1} > \tau_j$. The Dini derivative of $\mathfrak{V}_j$ is

$$D^+ \mathfrak{V}_j = \ell \zeta'^{\tau_j} (P_j \Psi_j + \Psi_j^T P_j) \zeta'_j + 2\ell(q_j + Q_j) L_j P_j \zeta'_j$$

$$\begin{aligned} &+ 2\delta B_{r_j+1}^T P_j \zeta'_j + 2\varpi'_j \mathfrak{B}_j^T P_j \zeta'_j + (\mu_j - v_j) [\ell \theta \tau_j \\ &- \ell(\theta + \kappa) \zeta'^2_{j,1} - \ell(\theta + \kappa)(\ell^{2\tau_j} v_j^2 + 2\ell^{\tau_j} v_j \zeta'_{j,1})] \\ &+ 2\mu_j \zeta'_{j,1} (\ell \zeta'_{j,2} - \ell L_{j,1} \zeta'_{j,1} + \ell L_{j,1} q_1 + \ell L_{j,1} Q_1). \end{aligned} \quad (25)$$

Similarly, there exist $\iota_{j,0}, \iota_{j,1}, \iota_{j,2}, \iota_{j,3}$ such that

$$\begin{aligned} D^+ \mathfrak{V}_j &< -\ell \iota_{j,0} \|\zeta'_j\|^2 + \ell^{\tau_j+1} \iota_{j,1} \|v\|_\infty \|\zeta'_j\| \\ &\quad + \iota_{j,2} \|e^{(0 \sim n-1)}\|^2 + \ell^{2\tau_j+1} \iota_{j,3} \|v\|_\infty^2, \end{aligned} \quad (26)$$

Based on (25) and $\|\varepsilon'\| \leq a'_g \|\zeta'\|, a'_g > 0$, it turns out that

$$\begin{aligned} D^+ V'_1 &< -\ell \min_{j \in [m]} \iota_{j,0} \|\zeta'\|^2 + m \ell^{\tau_j+1} \max_{j \in [m]} \iota_{j,1} \|v\|_\infty \|\zeta'\| \\ &\quad + m \ell^{2\tau_j+1} \max_{j \in [m]} \iota_{j,3} \|v\|_\infty^2 + m \max_{j \in [m]} \iota_{j,2} \|e^{(0 \sim n-1)}\|^2 \\ &\quad - \kappa \|e^{(0 \sim n-1)}\|^2 + 2a'_g \|A_{0 \sim n-1}\| \|B_{r_j}^T \mathcal{P}\| \|\zeta'\| \|e^{(0 \sim n-1)}\| \\ &\quad + 2(a'_1 + a'_2 \|\zeta'\| + a'_3 \|e^{(0 \sim n-1)}\|) \|B_{r_j}^T \mathcal{P}\| \|e^{(0 \sim n-1)}\| \\ &< -\ell(\min_{j \in [m]} \iota_{j,0} - \frac{1}{4}) \|\zeta'\|^2 + m \ell^{n+1} \max_{j \in [m]} \iota_{j,1} \|v\|_\infty \|\zeta'\| \\ &\quad - (\kappa - b'_3) \|e^{(0 \sim n-1)}\|^2 + b'_4 \|e^{(0 \sim n-1)}\|^2 \\ &\quad + m \ell^{2n+1} \max_{j \in [m]} \iota_{j,3} \|v\|_\infty^2, \end{aligned} \quad (27)$$

where $b'_1 = 2a'_g \|A_{0 \sim n-1}\| \|B_{r_j}^T \mathcal{P}\|, b'_2 = 2a'_2 \|B_{r_j}^T \mathcal{P}\|, b'_3 = (2b_1^2 + 2b_2^2)/\ell + m \max_{j \in [m]} \iota_{j,2} + 2a'_3 \|B_{r_j}^T \mathcal{P}\|, b'_4 = 2a'_1 \|B_{r_j}^T \mathcal{P}\|$. If the parameters satisfy $\min_{j \in [m]} \iota_{j,0} - 0.25 > 0, \kappa - b'_3 > 0$, the error $[\zeta'^{\tau_j}, e^{(0 \sim n-1)/\tau_j}]^T$ is UUB, and the error bound can be adjusted small enough via the appropriate parameters ℓ, κ . However, the performance of the proposed observer-saturated FAS controller is still affected by the noise item $\ell^n \|v\|_\infty$.

It follows that the estimation error ζ'_j converges exponentially, i.e., $\|\zeta'_j\| \leq \max\{\lambda'_j e^{-\alpha'_j \ell t}, \Lambda'_j / \ell + \ell^{\tau_j} \beta'_j \|v\|_\infty\}$, where $\lambda'_j, \alpha'_j, \Lambda'_j, \beta'_j, j \in [m]$ are positive scalars. Thus, the proof is completed. ■

Remark 2. Since measurement noise cannot be eliminated, the stability conclusions under the two controllers (9) and (19) are consistent. Despite the agreement between Lemma 5 and Lemma 1, the observer-saturated FAS structure has better suppression performance on the noise item $\ell^n \|v\|_\infty$. The error feedback item $\text{Sat}_{\sqrt{\tau_j}}(\ell^{\tau_j}(\xi_{j,1} + v_j - \hat{\xi}'_{j,1}))$ in (17) adopts the saturation suppression strategy, i.e., the error caused by $\ell^n \|v\|_\infty$ appearing at a certain moment will be quickly suppressed.

4.2. MFTC redesign with multiple faults

Following Section 3.2, there are some conclusions.

Lemma 6 (Cai et al., 2024). There exist sensor faults in the set I if

$$\|\mathfrak{R}'_I\| > \overline{\text{Lip}}(\gamma_I \circ \text{INV}(\gamma_I) - \mathcal{I})\gamma', \quad (28)$$

Conversely, if (28) fails in the 2d-redundant observable system (1) with no more than d sensor faults, the state observation, actuator fault estimation and sensor fault estimations are designed as

$$\hat{x}'_I \triangleq \hat{x}'_I^{(0 \sim n-1)'} = \text{INV}(\gamma_I)(\hat{\xi}'_I), \hat{\delta}'_I = \sum_{j \in I} \hat{\delta}'_j / (m - d),$$

$$\hat{s}'_{j,I} = y_j - H_j(\hat{x}'_I^{(0 \sim n-1)'}), j \in [m]. \quad (29)$$

Thus, there exist positive scalars $t'_3, \mathcal{M}'_1, \mathcal{M}'_2$ such that

$$\|x^{(0 \sim n-1)} - \hat{x}'_I\| \leq \frac{\mathcal{M}'_1}{\ell^2} + \ell^{n-1} \mathcal{M}'_2 \|v\|_\infty, t > t'_3. \quad (30)$$

As a result, the dynamical sensor set $\sigma(t)$ is updated as the next set $\sigma(t^+)$ if

$$\|\mathfrak{R}'_{\sigma(t)}\| > \overline{\text{Lip}}(\gamma_{\sigma(t)} \circ \text{INV}(\gamma_{\sigma(t)}) - \mathcal{I})\gamma'. \quad (31)$$

Under m observers (17), the novel MFTC is designed as

$$\begin{aligned}\hat{x}_{\sigma(t)}^{(0 \sim n-1)'} &= \text{INV}(\mathcal{Y}_{\sigma(t)})(\hat{x}'_{\sigma(t)}), \hat{\delta}'_{\sigma(t)} = \sum_{j \in \sigma(t)} \hat{\delta}'_j / (m - d), \\ \hat{s}'_{j, \sigma(t)} &= y_j - H_j(\hat{x}_{\sigma(t)}^{(0 \sim n-1)'})', j \in [m], \\ u'_2 &= -G^{-1}(\hat{x}_{\sigma(t)}^{(0 \sim n-1)'})[A_{0 \sim n-1}(\hat{x}_{\sigma(t)}^{(0 \sim n-1)'}) - x_r^{(0 \sim n-1)} \\ &\quad + F(\hat{x}_{\sigma(t)}^{(0 \sim n-1)'}) + \hat{\delta}'_{\sigma(t)} - x_r^{(n)}],\end{aligned}\quad (32)$$

where $\hat{x}_{\sigma(t)}^{(0 \sim n-1)'}$, $\hat{\delta}'_{\sigma(t)}$, $\hat{s}'_{j, \sigma(t)}$ are the corresponding estimations under the dynamical set $\sigma(t)$.

Theorem 1. For the $2d$ -redundant observable system (1) with multiple faults under the MFTC (32), the overall error is UUB. Thus, there exist positive scalars $t'_4, t'_5, M'_k (k = 7, 8, \dots, 14)$ such that

$$\begin{aligned}\|x^{(0 \sim n-1)} - \hat{x}_{\sigma(t)}^{(0 \sim n-1)'}\| &\leq M'_7/\ell^2 + \ell^{n-1}M'_8\|v\|_\infty, t > t'_4, \\ \|\delta - \hat{\delta}'_{\sigma(t)}\| &\leq M'_9/\ell + \ell^n M'_{10}\|v\|_\infty, t > t'_4, \\ \|s_j - \hat{s}'_{j, \sigma(t)}\| &\leq M'_{11}/\ell^2 + \ell^{n-1}M'_{12}\|v\|_\infty, j \in [m], t > t'_4, \\ \|e^{(0 \sim n-1)}\| &\leq M'_{13}/\ell + \ell^n M'_{14}\|v\|_\infty, t > t'_5.\end{aligned}\quad (33)$$

Proof. The proof is similar to Lemma 5. ■

4.3. Performance analysis in linear systems

The detection conditions (11) and (28) are the key to MFTC design, and the observation performance is the core of fault detection. Consider a linear structure:

$$\begin{aligned}\dot{x}^{(n)} &= Ax^{(0 \sim n-1)} + Bu + \delta, \\ \eta &= x + v_0,\end{aligned}\quad (34)$$

where $x, u, \eta \in \mathbb{R}$ are the system state, the control input and the measurement output, respectively. $A \in \mathbb{R}^{1 \times n}, B \in \mathbb{R}$ are two matrices. v_0 is the measurement noise, and δ is assumed to be a constant fault.

Let $\mathcal{X} = [x_1, x_2, \dots, x_{n+1}]^T = [x, \dot{x}, \dots, \dot{x}^{(n-1)}, \delta]^T$, and the observer based on Section 3 is designed as

$$\begin{aligned}\hat{x}_k &= \hat{x}_{k+1} + \ell^k \mathcal{L}_k(\eta - \hat{x}_1), k \in [n-1], \\ \hat{x}_n &= \mathcal{A}\hat{x}^{(0 \sim n-1)} + \hat{x}_{n+1} + \ell^n \mathcal{L}_n(\eta - \hat{x}_1) + Bu, \\ \hat{x}_{n+1} &= \ell^{n+1} \mathcal{L}_{n+1}(\eta - \hat{x}_1),\end{aligned}\quad (35)$$

where $\mathcal{L} = [\mathcal{L}_1, \dots, \mathcal{L}_{n+1}]^T$ can shape a Hurwitz matrix

$$\Psi = \begin{bmatrix} -\mathcal{L} & \\ -I_n & \mathbf{0}_{n-1} \end{bmatrix}^T. \quad (36)$$

Define the estimation error $\epsilon = [\epsilon_1, \epsilon_2, \dots, \epsilon_n, \epsilon_{n+1}]^T = [\ell^n(x_1 - \hat{x}_1), \ell^{n-1}(x_2 - \hat{x}_2), \dots, x_{n+1} - \hat{x}_{n+1}]^T$, and the derivative of ϵ is

$$\dot{\epsilon} = \ell \Psi_\epsilon \epsilon - \ell^{n+1} \mathcal{L} v_0, \quad (37)$$

where Ψ_ϵ is defined as

$$\Psi_\epsilon = \begin{bmatrix} -\mathcal{L}_1 & 1 & 0 & \dots & 0 \\ -\mathcal{L}_2 & 0 & 1 & \dots & 0 \\ \vdots & & & \ddots & \\ -\mathcal{L}_n + \frac{\mathcal{A}_1}{\ell^n} & \frac{\mathcal{A}_2}{\ell^{n-1}} & \frac{\mathcal{A}_3}{\ell^{n-2}} & \dots & 1 \\ -\mathcal{L}_{n+1} & 0 & 0 & \dots & 0 \end{bmatrix}, \quad (38)$$

and the i th element $\mathcal{A}$ is $\mathcal{A}_i$.

Accordingly, the observer based on Section 4 is

$$\begin{aligned}\hat{x}'_k &= \hat{x}'_{k+1} + \mathcal{L}_k \text{Sat}_{\sqrt{\tau}}(\vartheta) / \ell^{n-k}, k \in [n-1], \\ \hat{x}'_n &= \mathcal{A}\hat{x}'^{(0 \sim n-1)'} + \hat{x}'_{n+1} + Bu + \mathcal{L}_n \text{Sat}_{\sqrt{\tau}}(\vartheta),\end{aligned}$$

$$\begin{aligned}\dot{\hat{x}}_{n+1} &= \ell \mathcal{L}_{n+1} \text{Sat}_{\sqrt{\tau}}(\vartheta), \\ \dot{\tau} &= -\ell \theta \tau + \ell(\theta + \kappa) \vartheta^2, \tau \geq 0,\end{aligned}\quad (39)$$

where $\vartheta = \ell^n(\eta - \hat{x}'_1)$. Define the estimation error

$$\begin{aligned}\epsilon' &= [\epsilon'_1, \epsilon'_2, \dots, \epsilon'_n, \epsilon'_{n+1}]^T \\ &= [\ell^n(x_1 - \hat{x}'_1), \ell^{n-1}(x_2 - \hat{x}'_2), \dots, x_{n+1} - \hat{x}'_{n+1}]^T,\end{aligned}\quad (40)$$

and the derivative of ϵ' is

$$\begin{aligned}\dot{\epsilon}' &= \ell \Psi'_\epsilon \epsilon' + \ell \mathcal{L} \text{Sat}_{\sqrt{\tau}}(\epsilon'_1 + \ell^n v_0), \\ \dot{\tau} &= -\ell \theta \tau + \ell(\theta + \kappa)(\epsilon'_1 + \ell^n v_0)^2,\end{aligned}\quad (41)$$

where $\Psi'_\epsilon = \Psi'_\epsilon + \mathcal{L}$.

To compare the noise suppression performance of the two observers (35) and (39), the noise is selected as

$$v_0 = \mathcal{D}_p(t) \triangleq \begin{cases} 1/p, & 0 < t \leq p, \\ 0, & t \geq p, \end{cases} \quad (42)$$

where $\mathcal{D}_p(t), p \rightarrow 0^+$ denotes the impulse function. Then,

$$\|\epsilon'(p)\| / \|\epsilon(p)\|, p \rightarrow 0^+ \quad (43)$$

is analyzed for the performance comparison.

Lemma 7 (Astolfi et al., 2021). For the linear HOFAS model (34) under the two observers (35) and (39), if $\epsilon'(0) = \epsilon(0), \tau(0) = 0, v_0 = \mathcal{D}_p(t)$, the following condition holds:

$$\frac{\|\epsilon'(p)\|}{\|\epsilon(p)\|} \leq p \ell(\theta + \kappa) \frac{1 + o(p)}{1 - o(p)}, p \rightarrow 0^+, \quad (44)$$

where $o(p)$ denotes the infinitesimal term.

Remark 3. Lemma 7 means the saturated observer has better suppression performance on measurement noise. Especially when the noise has impulsive characteristics, the observation error of saturation redesign is smaller, i.e., γ' is smaller than γ . The fault detection logic (28) is less impacted by the noise item $\ell^n\|v\|_\infty$, thereby the observer-saturated MFTC technique outperforms.

Remark 4. The saturated design is inspired by Alessandri and Zaccarian (2018), Astolfi et al. (2021), and Lemma 7 is to supplement the saturation observer idea within the HOFAS theory. There are main differences: (a) The process of proving stability is different. The MFTC frameworks have taken further consideration of issues related to reliability and safety. (b) (32) is a fusion of state observations under two types of faults, and the MFTC design lies in the output-dependent decomposition observable analysis in the multi-output scenario. Compared with Kim et al. (2019), Lee et al. (2019), the MFTC redesign incorporates an actuator fault estimator, sensor fault estimator and an FAS fault-tolerant controller. Additionally, the proposed MFTC redesign is the more advanced version of our previous work (Cai et al., 2024).

5. Simulation

To illustrate the feasibility of the MFTC strategy, a nonlinear system is chosen as

$$\begin{aligned}x^{(3)} &= 3 - 3\ddot{x} - 2\dot{x} - \sin(\ddot{x} + 2\dot{x} - 2) + u + \delta(t), \\ y_j &= x + s_j(t) + v_j, j \in [3], \\ y_4 &= 2\dot{x} + 4x - 2 + s_4(t) + v_4,\end{aligned}\quad (45)$$

where $\delta = -\rho(t)u + f_a(t)$ is the actuator fault signal. The reference signal is set as $x_r = \sin t$. The relative degrees of four measurements are $r_1 = 3, r_2 = 3, r_3 = 2, r_4 = 2$, and the third-order system (45) is 2-redundant observable. The actuator fault

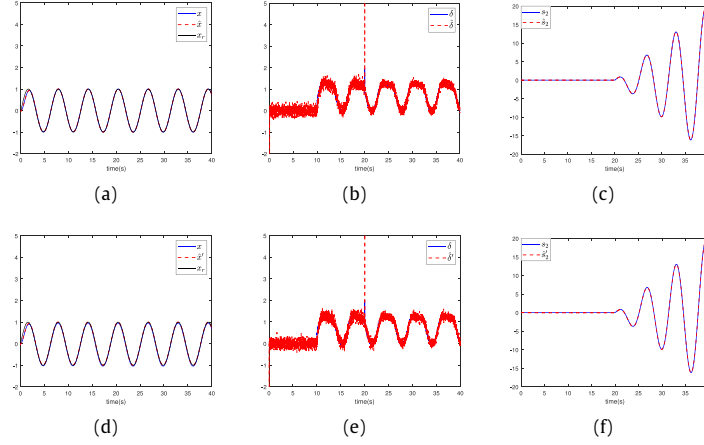


Fig. 1. Case 1: (a)–(c) The signals under the observer-extended MFTC; (d)–(f) The signals under the observer-saturated MFTC.

fails with the amplitude $\delta = -0.2u + \sin(t - 10)$ when $t > 10$ s. The second sensor fails at $t = 20$ s. The inverse mappings are

$$\begin{aligned} \gamma_{\{2,3,4\}}^{-1} &= [\xi_{2,1}, \xi_{3,1}, 0.5\xi_{4,2} - 2\xi_{3,1}]^T, \\ \gamma_{\{1,3,4\}}^{-1} &= [0.25\xi_{4,1} - 0.5\xi_{3,1} + 0.5, \xi_{1,2}, 0.5\xi_{4,2} - 2\xi_{1,2}]^T, \\ \gamma_{\{1,2,4\}}^{-1} &= [\xi_{1,1}, \xi_{2,2}, 0.5\xi_{4,2} - 2\xi_{1,2}]^T, \\ \gamma_{\{1,2,3\}}^{-1} &= [\xi_{1,1}, \xi_{2,2}, \xi_{3,2}]^T. \end{aligned} \quad (46)$$

In the following MFTC experiments, the compact set conditions are $x^{(0\sim 2)} \in \{\zeta \mid \zeta \in \mathbb{R}^3 : \|\zeta\|_\infty \leq 10\}$ and $\xi \in \{\zeta \mid \zeta \in \mathbb{R}^{10} : \|\zeta\|_\infty \leq 20\}$. The parameters are selected as $\ell = 20, \theta = 0.1, \kappa = 0.8, A^{0\sim 2} = [2, 4, 3], L_1 = L_2 = [4, 7, 6, 2], L_3 = L_4 = [2, 4, 3]$.

5.1. Case 1: The two MFTC experiments

In this case, the noise follows a uniform distribution over $[-0.001, 0.001]$. Fig. 1 gives the corresponding signal trajectories. Both MFTC frameworks can tolerate actuator faults and sensor faults effectively, illustrating Lemma 4 and Theorem 1. Higher-order states are more susceptible to noise. The actuator fault occurs at the highest-order position, and the oscillation cause of fault estimation comes from noise. The reason why the observer-saturated MFTC framework shows no obvious effect in noise suppression is that the deterministic structure is unable to completely eliminate noise. After extensive simulation analysis, it was found that the observer-saturated MFTC framework performs exceptionally well in the presence of impulse noise, which is displayed in Case 2.

5.2. Case 2: The noise suppression experiment

In this case, the noise is selected as impulse noise, and the pulses occur at $t = 30$ s. The key to noise suppression performance analysis is the state observation error because fault detection and fault tolerance are based on ideal state observations. Fig. 2 and Table 1 imply that the error resulting from noise in the second observer-saturated framework is smaller. It is obvious that in the moment of impulse noise, the saturated observer receives less impact and is more suitable for noisy systems. Although Case 1 illustrates the similar control performance of the two MFTC frameworks, Case 2 reveals that the observer-saturated MFTC framework has better noise suppression performance. After numerous experiments, the observer-saturated MFTC framework demonstrates superior performance in impulse-excitation-type noise, but its suppression effect on continuous-excitation-type noise is not significant.

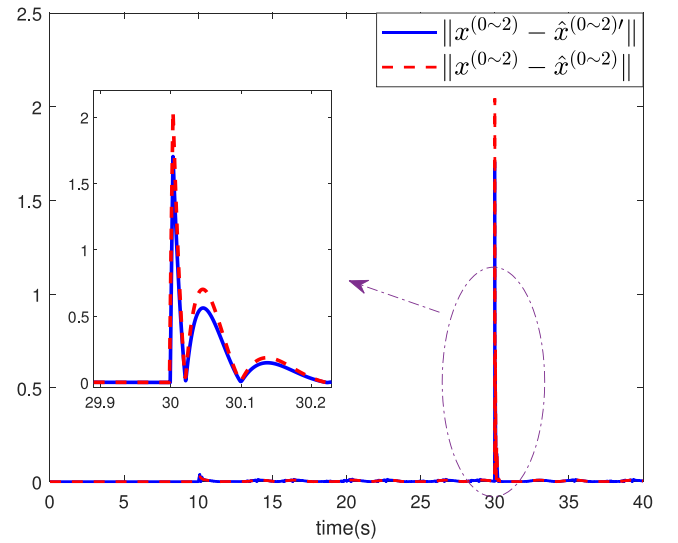


Fig. 2. Case 2: The observation error under two frameworks.

Table 1

The noise suppression comparison.

	1st framework	2nd framework
1st error peak	2.1	1.7
2nd error peak	1.1	0.9

6. Conclusion

In this paper, two MFTC frameworks based on FAS approaches are established. The paper progresses from the FTC structure that only tolerates actuator faults to the MFTC structure that can tolerate both actuator and sensor faults simultaneously. Furthermore, the traditional MFTC framework has been improved to a novel MFTC framework with saturated observations.

The key contributions lie in three aspects: (a) From the perspective of HOFAS theory, the MFTC structures can maintain the characteristics of controller parameterization, and the controller design process is general and standard. (b) From the perspective of FTC theory, the MFTC structures cannot only quickly compensate actuator faults, but also tolerate sudden and catastrophic sensor faults. (c) From the perspective of noise suppression, the

proposed observer-saturated MFTC structure has better performance. The structures can serve as a guide for sensor arrangement regulations except for an important supplement to the FTC theory. In the future, the FTC design for multiple-input-multiple-output HOFASs with unknown input matrices is more worthy of exploration, as this problem involves complex issues such as matrix identification (Ni et al., 2025).

References

- Alessandri, A., & Zaccarian, L. (2018). Stubborn state observers for linear time-invariant systems. *Automatica*, 88, 1–9.
- Astolfi, D., Alessandri, A., & Zaccarian, L. (2021). Stubborn and dead-zone redesign for nonlinear observers and filters. *IEEE Transactions on Automatic Control*, 66(2), 667–682.
- Cai, M., He, X., & Zhou, D. (2023a). An active fault tolerance framework for uncertain nonlinear high-order fully-actuated systems. *Automatica*, 152, Article 110969.
- Cai, M., He, X., & Zhou, D. (2023b). Low-power fault-tolerant control for nonideal high-order fully actuated systems. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 53(8), 4875–4887.
- Cai, M., He, X., & Zhou, D. (2024). Self-healing fault-tolerant control for high-order fully actuated systems against sensor faults: A redundancy framework. *IEEE Transactions on Cybernetics*, 54(4), 2628–2640.
- Chen, S., Wang, W., Fan, J., & Ji, Y. (2023). Impact angle constraint guidance law using fully-actuated system approach. *Aerospace Science and Technology*, 136, Article 108220.
- Cui, K.-X., Duan, G.-R., & Hou, M.-Z. (2024). Discrete-time model reference tracking control for a class of combined spacecraft: A high-order fully actuated system approach. *IEEE Transactions on Automation Science and Engineering*, 21(4), 6966–6977.
- Dong, R., Hua, C., Li, K., & Meng, R. (2023). Adaptive fault-tolerant control for high-order fully actuated system with full-state constraints. *Journal of the Franklin Institute*, 360(12), 8062–8074.
- Duan, G. (2021). High-order fully actuated system approaches: Part I. Models and basic procedure. *International Journal of Systems Science*, 52(2), 422–435.
- Duan, G.-R. (2024). Fully actuated system approach for control: An overview. *IEEE Transactions on Cybernetics*, 54(12), 7285–7306.
- Gauthier, J., Hammouri, H., & Othman, S. (1992). A simple observer for nonlinear systems applications to bioreactors. *IEEE Transactions on Automatic Control*, 37(6), 875–880.
- Gong, M., Sheng, L., & Zhou, D. (2024). Robust fault-tolerant stabilisation of uncertain high-order fully actuated systems with actuator faults. *International Journal of Systems Science*, 55(12), 2518–2530.
- Isidori, A. (1995). *Nonlinear control systems* (3rd ed.). Berlin, Germany: Springer-Verlag.
- Khalil, H. K. (2017). Extended high-gain observers as disturbance estimators. *SICE Journal of Control, Measurement, and System Integration*, 10(3), 125–134.
- Kim, J., Lee, C., Shim, H., Eun, Y., & Seo, J. H. (2019). Detection of sensor attack and resilient state estimation for uniformly observable nonlinear systems having redundant sensors. *IEEE Transactions on Automatic Control*, 64(3), 1162–1169.
- Lee, C., Shim, H., & Eun, Y. (2019). On redundant observability: From security index to attack detection and resilient state estimation. *IEEE Transactions on Automatic Control*, 64(2), 775–782.
- Liu, G.-P. (2022). Coordination of networked nonlinear multi-agents using a high-order fully actuated predictive control strategy. *IEEE/CAA Journal of Automatica Sinica*, 9(4), 615–623.
- Liu, X., Chen, M., Zhou, D., & Sheng, L. (2024). Fault-tolerant control of stochastic high-order fully actuated systems. *IEEE Transactions on Cybernetics*, 54(5), 3225–3238.
- Liu, C., Liu, X., Wang, H., Zhou, Y., & Gao, C. (2023). Adaptive control for unknown HOF nonlinear systems without overparametrization. *International Journal of Robust and Nonlinear Control*, 33(6), 3640–3660.
- Lu, S., Tsakalis, K., & Chen, Y. (2023). Development and application of a novel high-order fully actuated system approach—Part I: 3-DOF quadrotor control. *IEEE Control Systems Letters*, 7, 1177–1182.
- Ma, Y., Zhang, K., & Jiang, B. (2023). Prescribed-time fault-tolerant control for fully actuated heterogeneous multiagent systems: A hierarchical design approach. *IEEE Transactions on Aerospace and Electronic Systems*, 59(5), 6624–6636.
- Meng, R., Hua, C., Li, K., & Ning, P. (2022). Adaptive event-triggered control for uncertain high-order fully actuated system. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 69(11), 4438–4442.
- Ni, J., Song, Y., Huang, X., & Wang, Y. (2025). Leader-follower consensus control with hierarchical prescribed performance for nonlinear multi-agent systems under reversing actuator faults. *Automatica*, 177, Article 112332.
- Ricardo Jr, J. A., & Santos, D. A. (2022). Smooth second-order sliding mode control for fully actuated multirotor aerial vehicles. *ISA Transactions*, 129, 169–178.
- Shim, H. (2000). *A passivity-based nonlinear observer and a semi-global separation principle* (Ph.D. thesis), Seoul National University.
- Zhang, D.-W., Liu, G.-P., & Cao, L. (2022). Coordinated control of high-order fully actuated multiagent systems and its application: A predictive control strategy. *IEEE/ASME Transactions on Mechatronics*, 27(6), 4362–4372.
- Zhao, Z., Zhang, J., Liu, Z., Li, H.-X., & Chen, C. P. (2024). Event-triggered adaptive neural fault-tolerant control for a 2-DOF helicopter system with prescribed performance. *Automatica*, 162, Article 111511.



Miao Cai received the B.Eng. degree in automation from Nanjing University of Science and Technology, Nanjing, China, in 2019, and the Ph.D. degree in control science and engineering from Tsinghua University, Beijing, China, in 2024.

He is currently a Lecturer with the School of Automation, Southeast University, Nanjing, China. His research interests include fully actuated system approaches, fault diagnosis and fault-tolerant control, as well as their applications.



Xiao He (SM'20-, IEEE) received the B.Eng. degree in information technology from Beijing Institute of Technology, Beijing, China, in 2004, and the Ph.D. degree in control science and engineering from Tsinghua University, Beijing, China, in 2010.

He is currently a Tenured Professor with the Department of Automation, Tsinghua University. He has authored more than 200 papers in refereed international journals. His research interests include fault diagnosis and fault-tolerant control, networked systems, cyber-physical systems, as well as their applications.

Dr. He is now a Full Member of Sigma Xi, the Scientific Research Society, a Senior Member of the Chinese Association of Automation, and a Senior Member of the IEEE. He is an Associate Editor of *IEEE Transactions on Neural Networks and Learning Systems* and *Control Engineering Practice*.



Donghua Zhou (SM'99–F'19, IEEE) received the B.Eng., M.Sci., and Ph.D. degrees in electrical engineering all from Shanghai Jiaotong University, China, in 1985, 1988, and 1990, respectively. He was an Alexander von Humboldt research fellow with the University of Duisburg-Essen, Germany, from 1995 to 1996, and a visiting scholar with Yale University, USA, from 2001 to 2002. He joined Tsinghua University, China, in 1996, where he was promoted to full professor in 1997 and was the head of the Department of Automation, from 2008 to 2015. He was a vice president of Shandong University of Science and Technology, China, from 2015 to 2023.

He is now a chief professor with Southeast University, China. He has authored and co-authored over 340 peer-reviewed international journal papers and 11 monographs in the areas of fault diagnosis, fault-tolerant control and operational safety evaluation.

Dr. Zhou is a fellow of IEEE, IET, CAA and AAIA, a member of IFAC TC on SAFEPROCESS. He was a vice-Chairman of Chinese Association of Automation (CAA) from 2013 to 2025, and was also the NOC Chair of the 6th IFAC Symposium on SAFEPROCESS 2006.